Online Algorithms for Learning and Control under Safety Constraints

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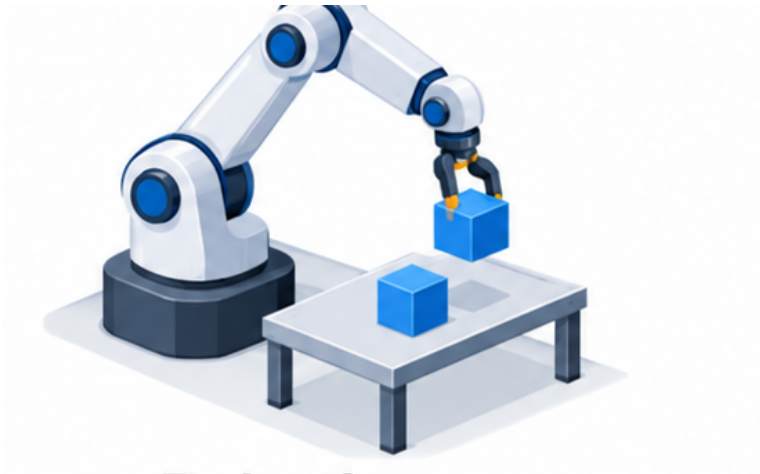
PhD Defense



Machine learning in modern engineered systems

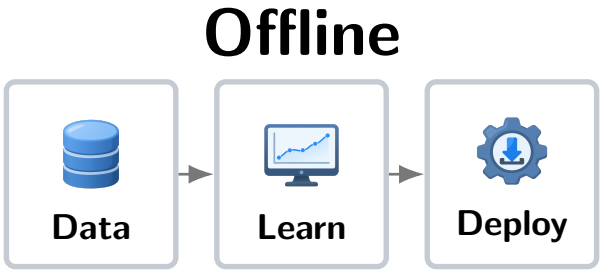


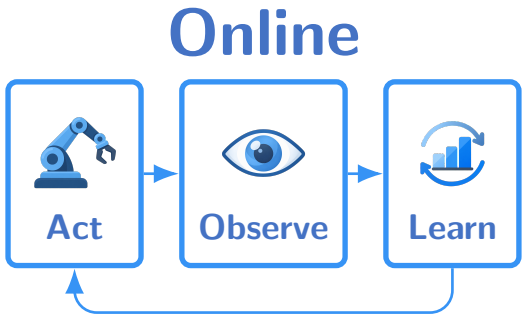






Why use online methods for learning?





Incomplete prior data
Prior data may be unavailable or omit important operating conditions.

Persistent model errors
Errors can accumulate due to model inaccuracies or system changes.

Actively gather data
Data is gathered interactively according to learning needs.

Continually improve model
New observations are used to correct inaccuracies and adapt to changes.

Online methods use feedback during operation to reduce uncertainty and improve decisions.

Online decisions are subject to safety constraints

Many real systems have constraints that must **always** be satisfied.



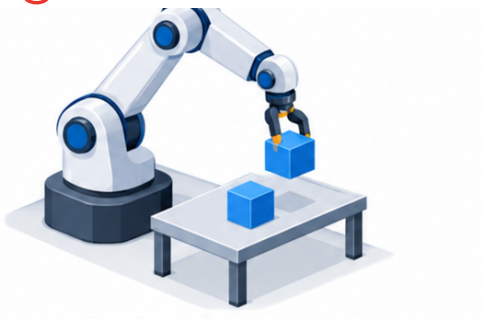
Electric power systems

Voltage and line-flow limits



Autonomous vehicles

Collision avoidance



Robotics

Workspace and actuator limits

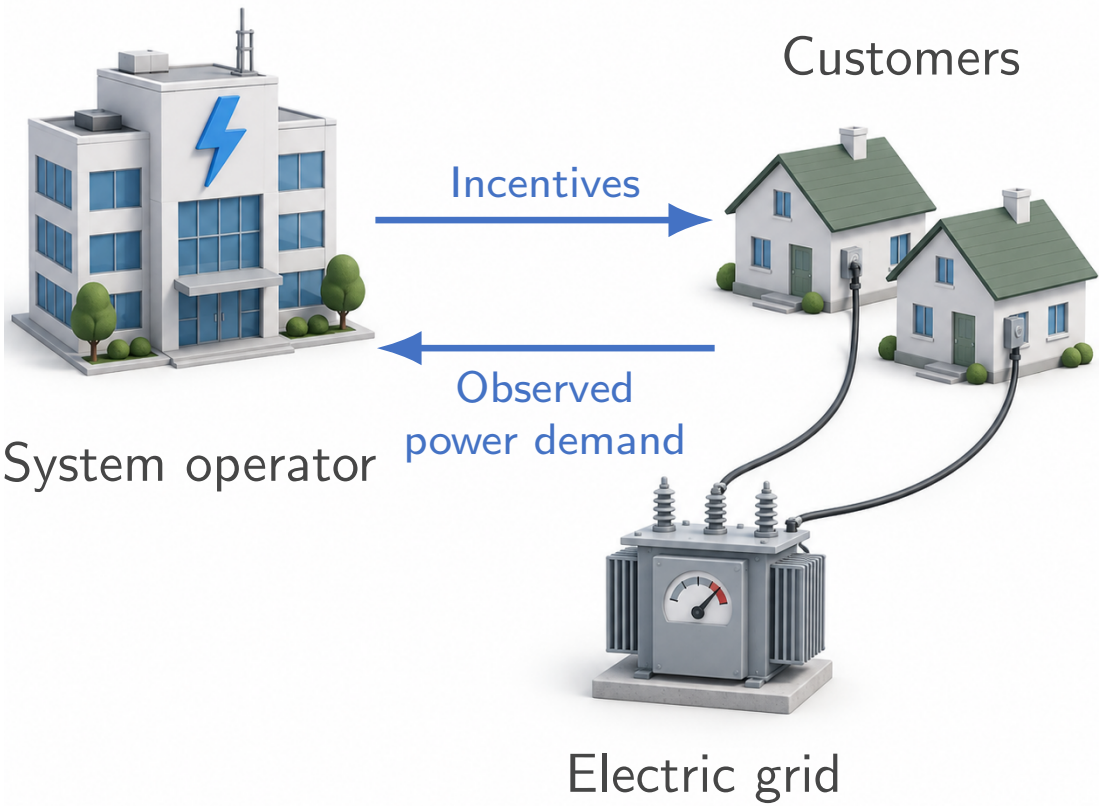


Smart buildings

Comfort and equipment limits

Online decisions must maintain constraints despite uncertainty.

Example: Online learning of demand response in the electric grid



Repeated decisions
Choose incentives, observe demand.

Uncertainty
Demand response and system behavior are only imperfectly known.

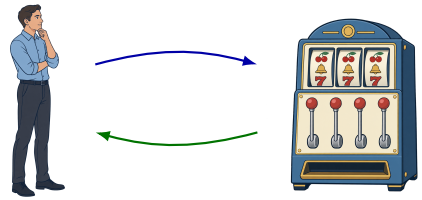
Operational constraints
Voltages, line flows, and device limits must remain acceptable.

Need to learn the best incentives through operation without violating constraints along the way.

Problem Settings for Safe Online Learning

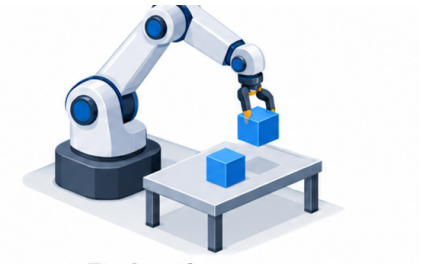
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   Learn unknown rewards and constraints from noisy feedback.



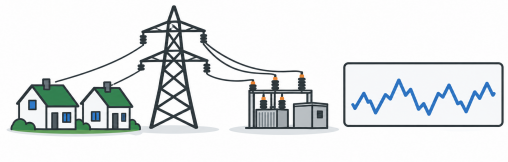
2. **Safe Online Control**

   Learn unknown dynamics while controlling the system.



3. **Safe Online Convex Optimization**

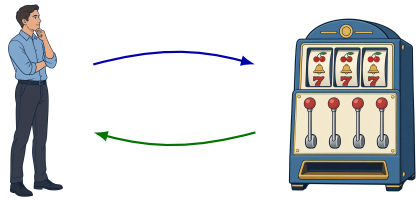
   Respond to changing costs while learning convex constraints.



Problem Settings for Safe Online Learning

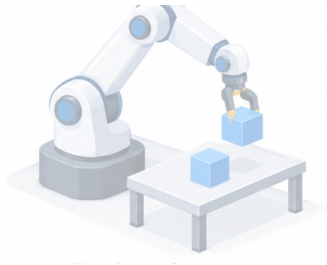
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   Learn unknown rewards and constraints from noisy feedback.



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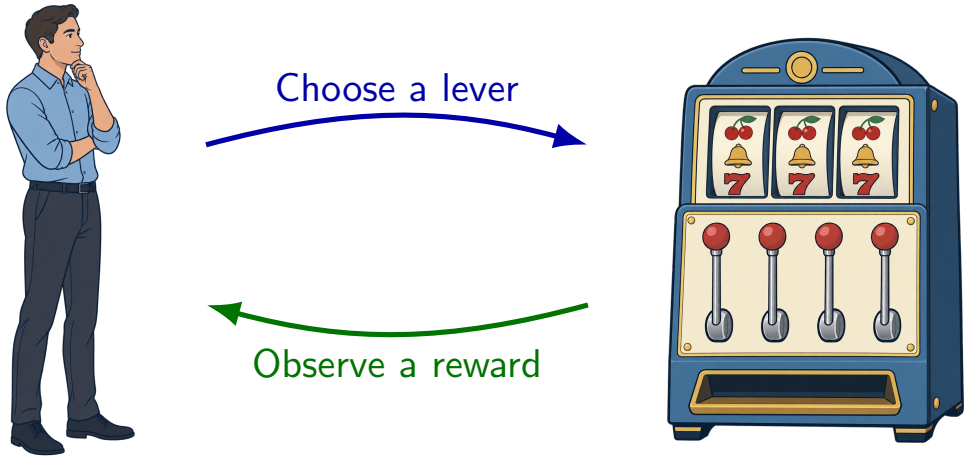


3. **Safe Online Convex Optimization**

   Respond to changing costs while learning convex constraints.

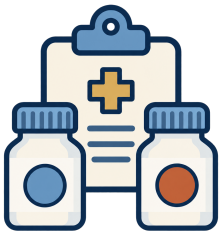


The Bandit Problem



"multi-armed bandit"

general model for decision-making under uncertainty



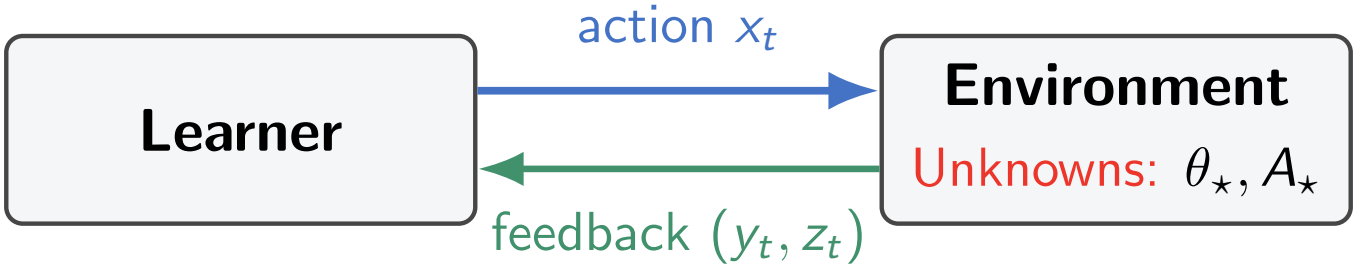


we consider a generalization with linear rewards and constraints

Safe Linear Bandits

Interaction at each round t:

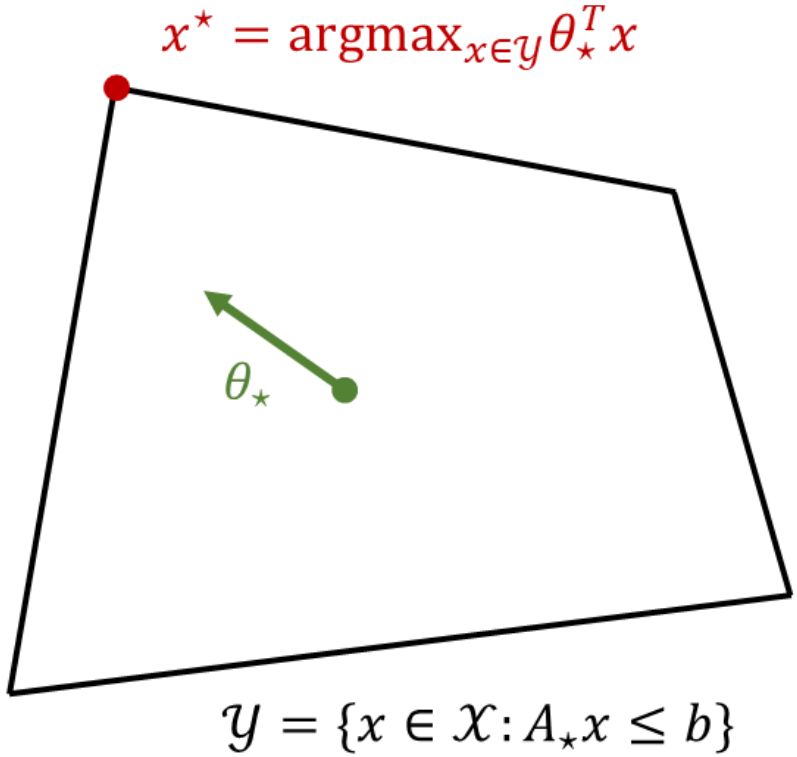
1. Choose an action: $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$.
2. Observe reward feedback: $y_t = \theta_\star^\top x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2)$.
3. Observe constraint feedback: $z_t = A_\star x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \Sigma)$.



Performance goals:

- Maximize cumulative reward: $\sum_{t=1}^{T} \theta_\star^\top x_t$.
- Satisfy the constraints: $A_\star x_t \leq b$ for every t.

Problem Geometry



How to **maximize reward** and **ensure constraints** with unknown $\theta_\star, A_\star$?

Learning from the decisions we make

Each round adds a new observation of both unknowns.

Round	Action	Reward observation	Constraint observation
1	x_1	$y_1 = \theta_\star^\top x_1 + \epsilon_1$	$z_1 = A_\star x_1 + \eta_1$
2	x_2	$y_2 = \theta_\star^\top x_2 + \epsilon_2$	$z_2 = A_\star x_2 + \eta_2$
3	x_3	$y_3 = \theta_\star^\top x_3 + \epsilon_3$	$z_3 = A_\star x_3 + \eta_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

At round t: use observations from rounds $1, \ldots, t-1$

Which parameters best fits the observed values?

Fitting the unknown linear functions

Reward estimate: fit a linear function to (x_s, y_s).

$$\widehat{\theta}_t \in \arg\min_{\theta} \left\{ \sum_{s=1}^{t-1} \underbrace{(y_s - \theta^\top x_s)^2}_{\text{squared prediction error}} + \underbrace{\lambda \|\theta\|_2^2}_{\text{regularization}} \right\}.$$

Constraint estimate: fit the observations (x_s, z_s) similarly.

$$\widehat{A}_t \in \arg\min_{A} \left\{ \sum_{s=1}^{t-1} \|z_s - A x_s\|_2^2 + \lambda \|A\|_F^2 \right\}.$$

The fitted parameters give best guess, but doesn't tell us about uncertainty.

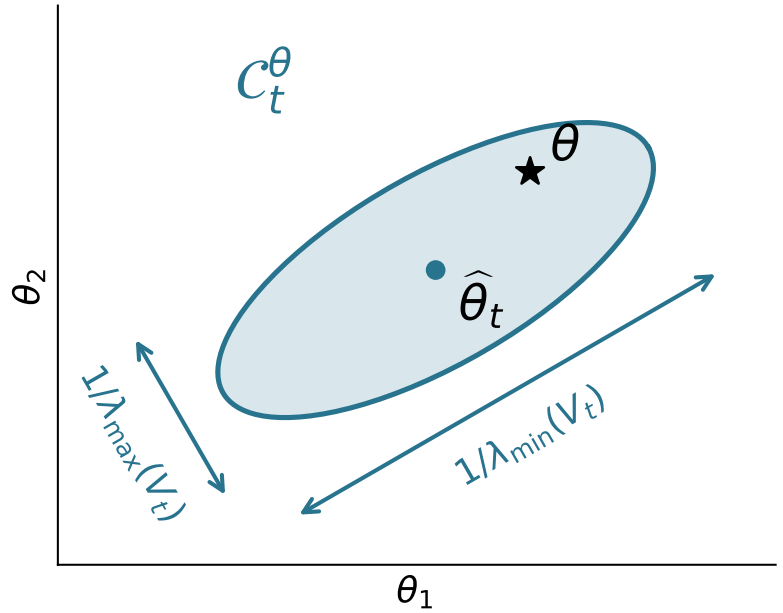
Confidence bounds on true parameters

Estimation error bounds[1](w.h.p.):

$$(\theta_\star - \widehat{\theta}_t)^\top V_t(\theta_\star - \widehat{\theta}_t) \leq \beta^2,$$
$$(a_{i,\star} - \hat{a}_{i,t})^\top V_t(a_{i,\star} - \hat{a}_{i,t}) \leq \beta^2, \forall i$$

Past actions determine tightness:

$$V_t = \lambda I + \sum_{s=1}^{t-1} x_s x_s^\top$$



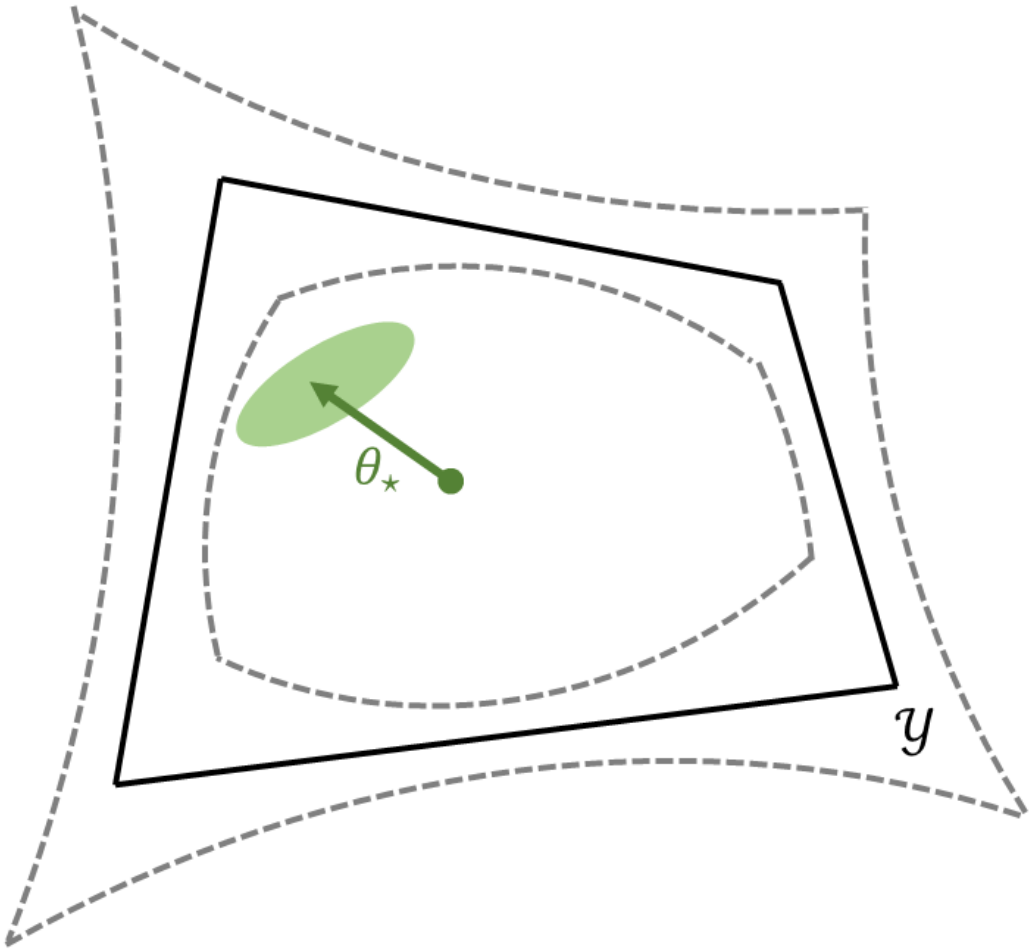
Confidence sets. It holds with high probability that:

$$\theta_\star \in C_t^\theta := \left\{\theta : (\theta - \widehat{\theta}_t)^\top V_t(\theta - \widehat{\theta}_t) \leq \beta^2\right\},$$
$$A_\star \in C_t^A := \left\{A : (a_i - \hat{a}_{i,t})^\top V_t(a_i - \hat{a}_{i,t}) \leq \beta^2, \forall i\right\}$$

[1] Abbasi-Yadkori et al. (2011)

Confidence bounds characterize problem uncertainty

Confidence bounds induce bounds on reward and constraints.



How do we use these uncertainty bounds?

Optimism in the Face of Uncertainty

"take the unknown to be as favorable as reasonable possible"

Aim to learn optimal:

$$x^\star = \arg\max_{x \in \mathcal{X}} \ \theta_\star^\top x$$
$$\text{s.t. } A_\star x \leq b$$

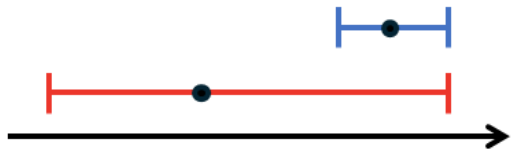
Take unknowns $A_\star, \theta_\star$ to be favorable as possible given uncertainty:

$$\tilde{x}_t = \arg\max_{x \in \mathcal{X}} \ \left(\max_{\theta \in \mathcal{C}_t^\theta} \theta^\top x \right)$$
$$\text{s.t. } Ax \leq b \text{ for some } A \in \mathcal{C}_t^A$$

By choosing $\tilde{x}_t$, we either get high reward, or informative feedback.

$\tilde{x}_t$ has high actual reward $\implies$ receive high reward

$\tilde{x}_t$ has high uncertainty $\implies$ feedback is informative

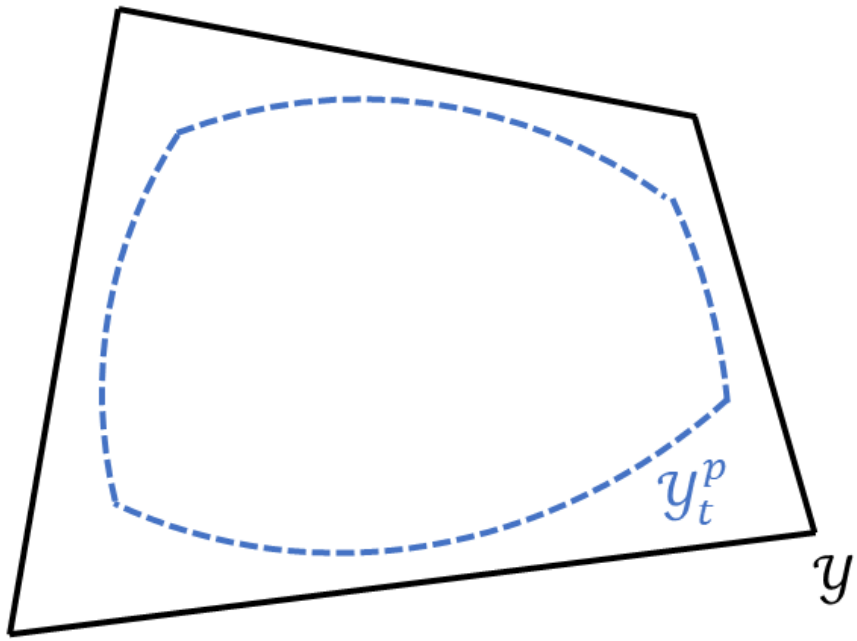


However, $\tilde{x}_t$ may violate the true constraints!

Pessimistic set ensures constraint satisfaction

Pessimistic set: smallest possible feasible set given uncertainty.

$$\mathcal{Y}_t^p = \left\{ x \in \mathcal{X} : Ax \leq b, \forall A \in \mathcal{C}_t^A \right\}$$



Pessimistic set is within true feasible set: $\boxed{\mathcal{Y}_t^p \subseteq \mathcal{Y}}$ w.h.p.

ROFUL Algorithm

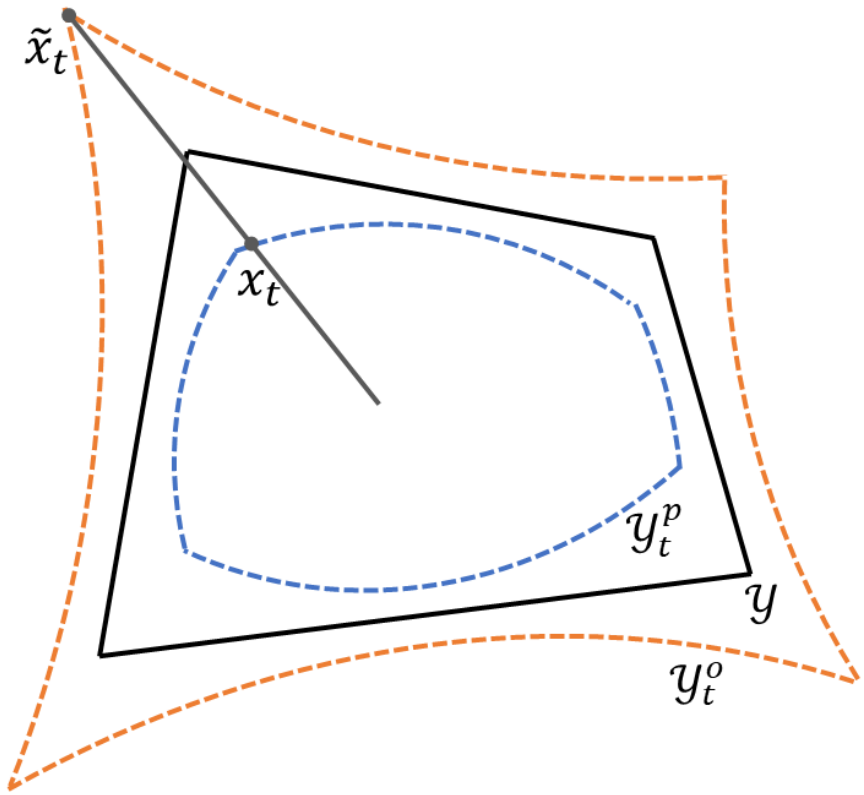
Step 1: **Optimistic target $\tilde{\mathbf{x}}_t$**

$$\tilde{x}_t = \arg\max_{x \in \mathcal{X}} \left(\max_{\theta \in \mathcal{C}_t^\theta} \theta^\top x \right)$$

$$\text{s.t. } Ax \leq b \text{ for some } A \in \mathcal{C}_t^A$$

Step 2: **Safe scaling**

$$\gamma_t = \max\{\gamma \in [0,1] : \gamma \widetilde{x}_t \in \mathcal{Y}_t^p\}$$
$$x_t = \gamma_t \widetilde{x}_t.$$



action is chosen from pessimistic set $\implies$ constraint satisfaction

Why use scaling to ensure constraint satisfaction?

Scaling depends on uncertainty at chosen action

Lemma. ROFUL algorithm ensures that:

$$\|x_t - \tilde{x}_t\| \lesssim \beta \|x_t\|_{V_t^{-1}} = \text{uncertainty at } x_t$$

Case 1: x_t is near to $\tilde{x}_t$

$\Longrightarrow$ **good performance** ✅

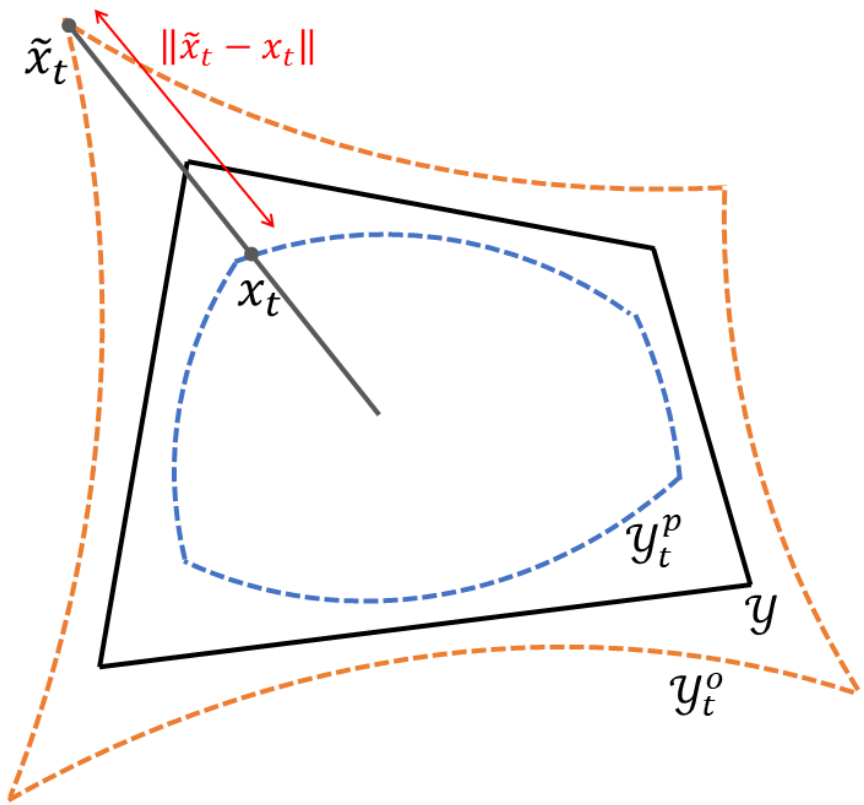
(since $\tilde{x}_t$ is optimistic)

Case 2: x_t is **not** near to $\tilde{x}_t$

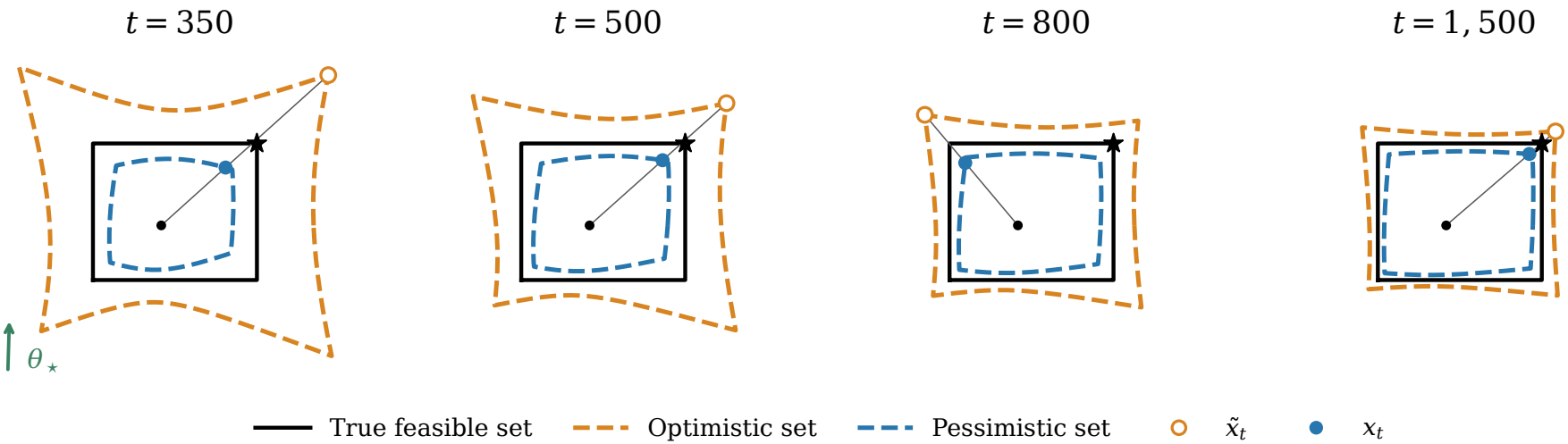
$\Longrightarrow$ high uncertainty at x_t

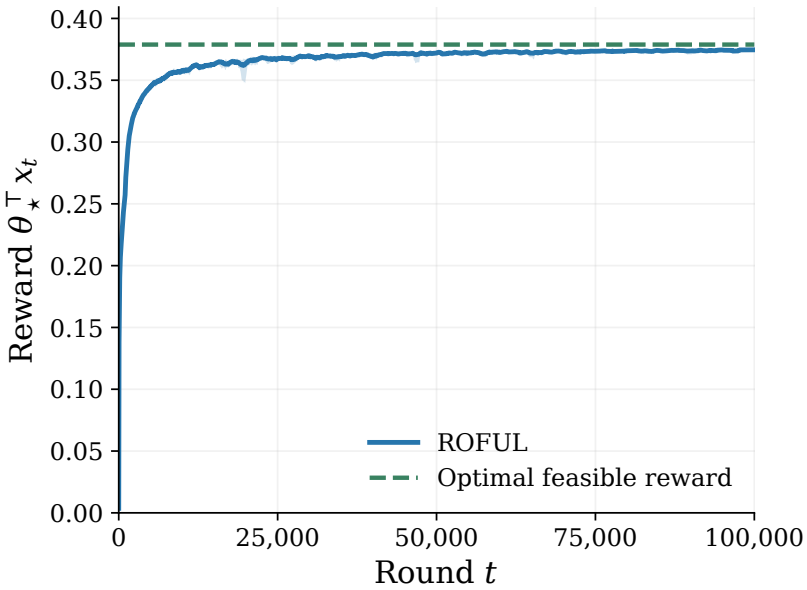
$\Longrightarrow$ **feedback is informative** ✅

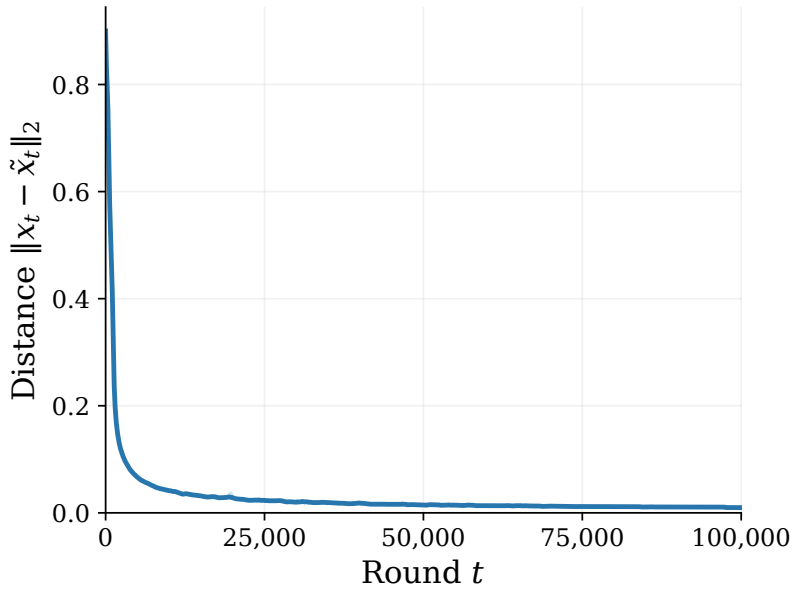
Win-win situation!



Numerical example







Theoretical Results

Regret measures gap between optimal reward and accumulated reward:

$$R_T = \sum_{t=1}^{T} \theta^\top x^\star - \sum_{t=1}^{T} \theta^\top x_t.$$

Theorem.[2] Assuming star-convex action set, Slater's condition, and sub-Gaussian noise, it holds with high probability

$$\boxed{R_T = \widetilde{O}(\sqrt{T})} \quad \text{and} \quad \boxed{Ax_t \leq b \quad \forall t \leq T.}$$

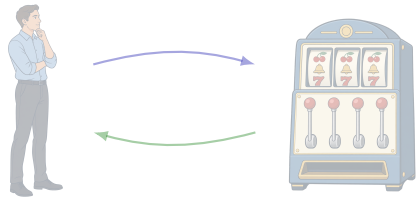
- **Novel algorithm** for safe linear bandits.
- Matches **state-of-the-art regret bounds** for safe linear bandits
- $\sqrt{T}$ regret is known to be **optimal**

[2] **S. Hutchinson**, B. Turan, and M. Alizadeh, "Directional Optimism for Safe Linear Bandits," AISTATS 2024

Problem Settings for Safe Online Learning

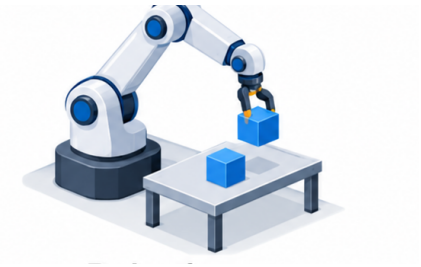
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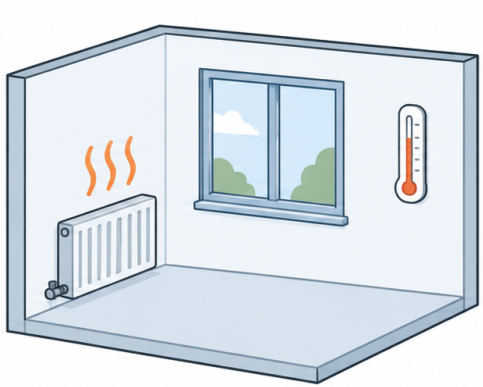
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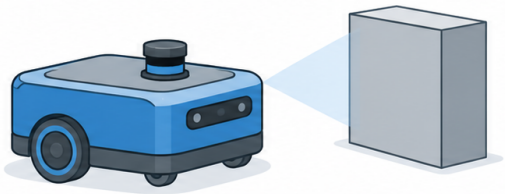
   Respond to changing costs while learning convex constraints.



Safe learning under dynamics

What if current actions impact future costs and constraints?







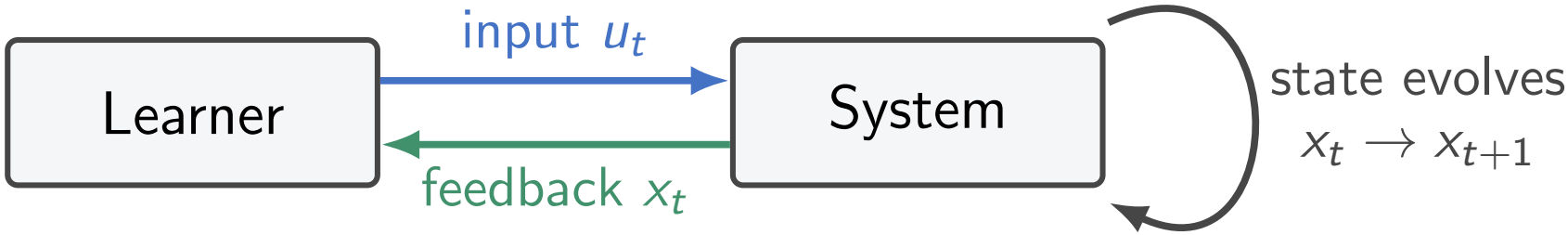
How can we **learn unknown dynamics**
while keeping the system **safe**?

Safe Online Control with Chance-constraints

Interaction at each time step t:

1. Observe state feedback x_t.
2. Choose input u_t.
3. State evolves according to unknown system:

$$x_{t+1} = A_\star x_t + B_\star u_t + w_t, \qquad w_t \overset{\text{iid}}{\sim} \mathcal{N}(0, W).$$



Satisfy chance constraints:

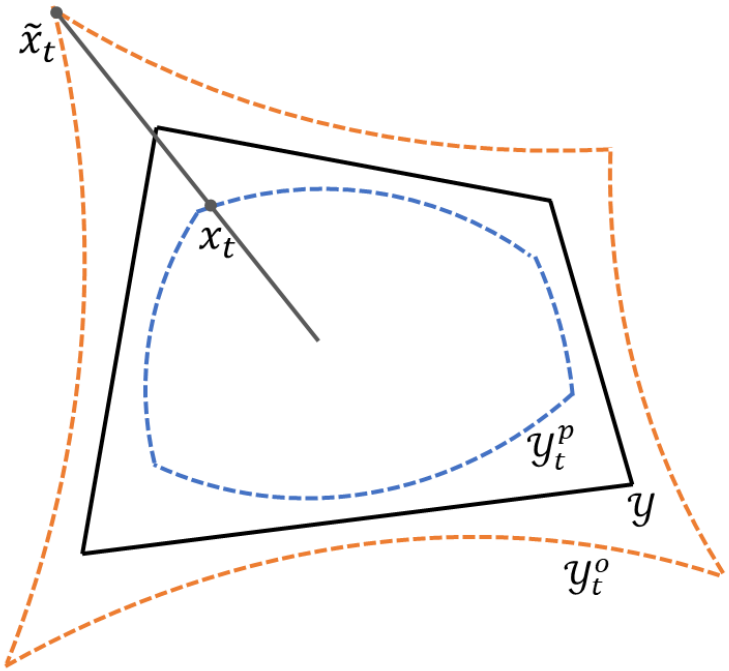
$$\mathbb{P}\left(\alpha_j^\top z_t \le \beta\right) \ge 1 - \delta, \quad z_t = \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \quad \forall t, j.$$

Minimize quadratic cost:

$$\sum_{t=1}^{T} \ell(x_t, u_t), \quad \ell(x, u) = x^\top Q x + u^\top R u$$

Optimism and safe scaling for control

Recall ROFUL



The same strategy for control

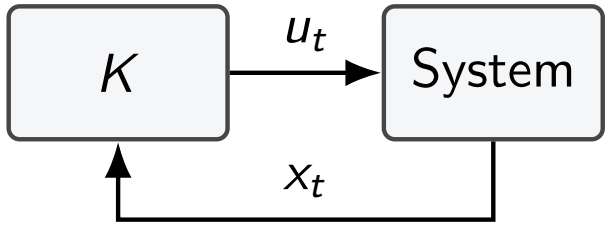
1. Find an optimistic design.
2. Move toward a safe baseline until safety can be certified.

What should we optimize and scale?

From linear controllers to covariance matrices

Consider **linear controllers:**

$$u_t = Kx_t, \qquad K = \text{feedback gain}$$



Directly optimizing K is **hard**:

cost function and chance-constraints are non-convex in K

Following prior work[3], we optimize **steady-state covariance** Σ, which determines steady-state distribution:

$$z_t = \begin{bmatrix} x_t \\ u_t \end{bmatrix} \xrightarrow{d} \mathcal{N}(0, \Sigma)$$

Expected cost is linear in Σ:

$$\mathbb{E}[\ell(x_t, u_t)] = \left\langle \left[\begin{smallmatrix} Q & 0 \\ 0 & R \end{smallmatrix}\right], \Sigma \right\rangle.$$

Chance-constraints are linear in Σ:

$$\mathbb{P}(\alpha_j^\top z_t \le \beta) \ge 1 - \delta \quad \iff \quad \langle \alpha_j \alpha_j^\top, \Sigma \rangle \le \beta^2 / \Phi^{-1}(1-\delta)^2.$$

[3] Cohen et al. (2018), Gattami (2010); Schildbach et al. (2015)

Controller design with known dynamics

If the system $A_\star, B_\star$ were known, could optimize covariance as:

$$\begin{aligned}
\min_{\Sigma \succeq 0} \quad & \langle \left[\begin{smallmatrix} Q & 0 \\ 0 & R \end{smallmatrix}\right], \Sigma \rangle && \text{expected cost} \\
\text{s.t.} \quad & \Sigma_{xx} = [A_\star \; B_\star] \Sigma [A_\star \; B_\star]^\top + W && \text{steady-state dynamics} \\
& \langle \alpha_j \alpha_j^\top, \Sigma \rangle \le \beta^2 / \Phi^{-1}(1-\delta)^2, \quad \forall j && \text{safety.}
\end{aligned}$$

A convex semidefinite program (SDP).

Recover a random linear controller from Σ:

$$\boxed{u_t = K x_t + v_t, \qquad v_t \sim \mathcal{N}(0, U)},$$

where gain and input covariance are calculated as:

$$K = \Sigma_{ux} \Sigma_{xx}^{-1}, \quad U = \Sigma_{uu} - \Sigma_{ux} \Sigma_{xx}^{-1} \Sigma_{xu}, \quad \Sigma = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xu} \\ \Sigma_{ux} & \Sigma_{uu} \end{bmatrix}$$

An optimistic target under unknown dynamics

Feedback provides regression data

$$x_{s+1} = \underbrace{[A_\star \; B_\star]}_{\Theta_\star} z_s + w_s.$$

Estimate $\widehat{\Theta}_k$ using least squares, with confidence bounds[4]:

$$\|\widehat{\Theta}_k \Sigma \widehat{\Theta}_k^\top - \Theta_\star \Sigma \Theta_\star^\top\| \le \eta \langle V_t^{-1}, \Sigma \rangle.$$

For *optimism*, relax dynamics $\Sigma_{xx} = \Theta_\star \Sigma \Theta_\star^\top + W$ accordingly:

$$\Sigma_{xx} \succeq \widehat{\Theta}_k \Sigma \widehat{\Theta}_k^\top + W - \eta \langle \bar{V}_k^{-1}, \Sigma \rangle I$$

Optimistic SDP

$$\Sigma_k^o = \arg\min_{\Sigma \succeq 0} \quad \langle \left[\begin{smallmatrix} Q & 0 \\ 0 & R \end{smallmatrix}\right], \Sigma \rangle$$

$$\text{s.t.} \quad \Sigma_{xx} \succeq \widehat{\Theta}_k \Sigma \widehat{\Theta}_k^\top + W - \eta \langle \bar{V}_k^{-1}, \Sigma \rangle I \quad \text{relaxed dynamics}$$

$$\langle \alpha_j \alpha_j^\top, \Sigma \rangle \le \beta^2 / \Phi^{-1}(1-\delta)^2, \quad \forall j.$$

[4] Cohen et al. (2019)

Certifying safety under uncertainty

A safe target covariance alone is insufficient.

Model uncertainty

The true dynamics differ from the relaxed dynamics.

Transient uncertainty

The system takes time to settle as the controller changes.

Lemma. Implementing policy Σ_t satisfying relaxed dynamics,

$$\mathrm{cov}(z_t) - \Sigma_t \precsim \underbrace{\eta\langle \bar{V}_k^{-1}, \Sigma\rangle}_{\text{model uncertainty}} + \underbrace{\frac{1}{\sqrt{T}}}_{\text{transient uncertainty}}$$

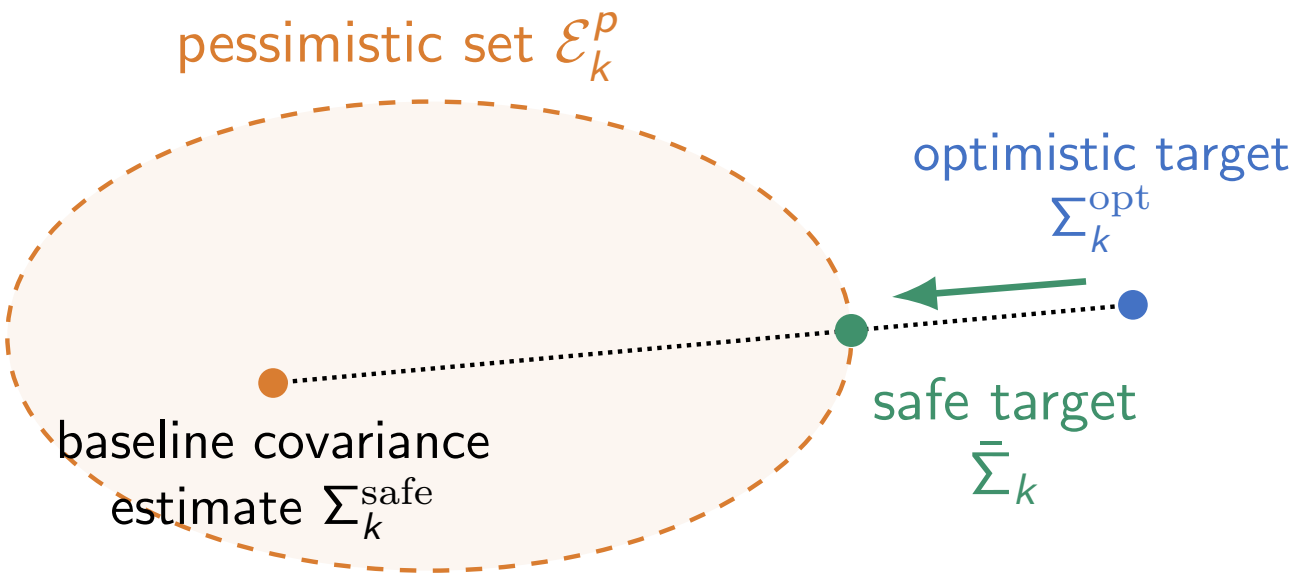
Pessimistic covariance set:

$$\mathcal{E}_k^p = \left\{\Sigma \succeq 0 : \langle \alpha_j \alpha_j^\top, \Sigma\rangle \leq \frac{\beta^2}{\Phi^{-1}(1-\delta)^2} - \eta\langle \bar{V}_k^{-1}, \Sigma\rangle - \frac{1}{\sqrt{T}},\ \forall j\right\}.$$

Safe scaling in covariance space

Choose the largest $\phi_k \in [0, 1]$ such that

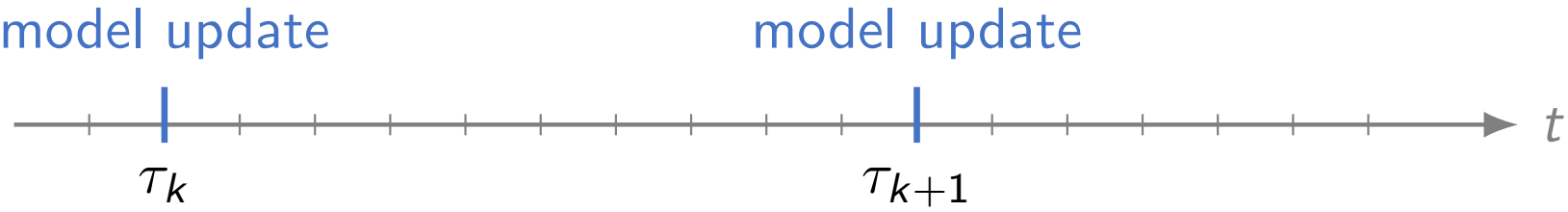
$$\bar{\Sigma}_k = \phi_k \Sigma_k^{\text{opt}} + (1 - \phi_k)\Sigma_k^{\text{safe}} \in \mathcal{E}_k^p.$$



As in ROFUL, the safety correction depends on uncertainty associated with the chosen design.

Implementing the controller over time

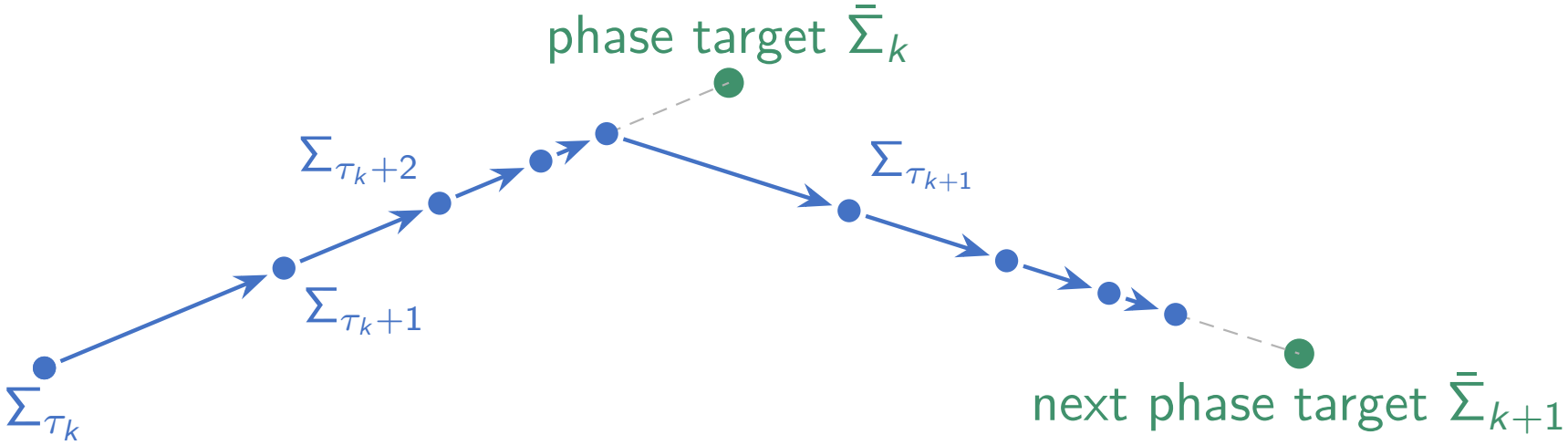
Only update model when enough new information arrives.



At a model update: ❶ estimate dynamics, ❷ optimistic SDP, ❸ scale.
This gives a safe phase target $\bar{\Sigma}_k$.

At each time step: move gradually toward that target:

$$\Sigma_t = (1-\zeta)\Sigma_{t-1} + \zeta\bar{\Sigma}_k, \qquad u_t \sim \mathcal{N}(K_t x_t, U_t).$$



Regret and constraint satisfaction

Regret against best safe linear controller $K^\star$:

$$R_T = \sum_{t=1}^{T} \ell(x_t, u_t) - \mathbb{E} \sum_{t=1}^{T} \ell(x_t^{K^\star}, u_t^{K^\star})$$

Theorem[5] With known safe-baseline, p.d. disturbance covariance, bounded parameters, sufficiently large T:

$$\boxed{R_T = \widetilde{O}(\sqrt{T})} \quad \text{with high probability.}$$

$$\boxed{\mathbb{P}(\alpha_j^\top z_t \leq \beta) \geq 1 - \delta} \qquad \forall t, j.$$

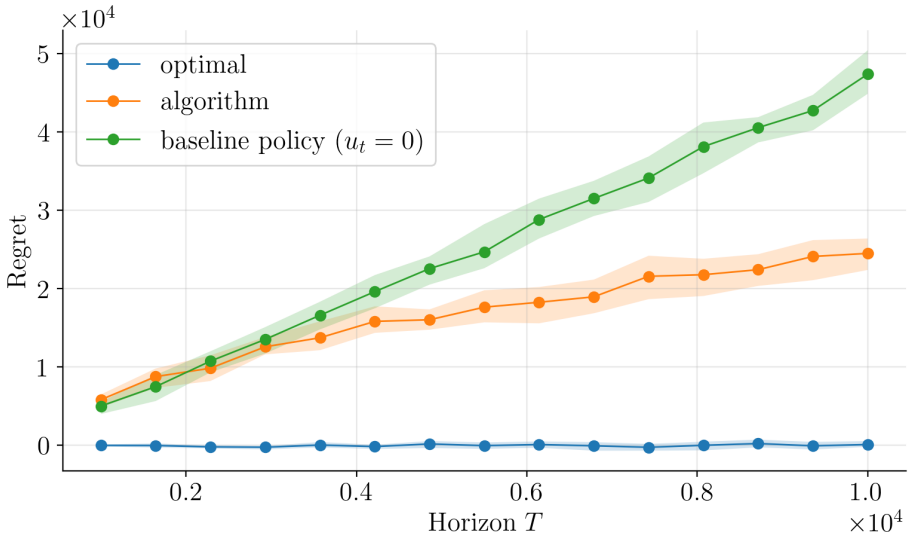
- Novel **covariance-based analysis**. Prior analysis is cost-to-go based.
- Regret bound is **optimal** in T.
- **First** $\widetilde{O}(\sqrt{T})$ regret for constrained linear quadratic regulator. Prior work has shown $\widetilde{O}(T^{2/3})$ under robust constraints.

[5] **S. Hutchinson**, N. Jiang, M. Alizadeh, "Rate-Optimal Regret for the Safe Learning-based Control of the Constrained Linear Quadratic Regulator," arXiv 2026

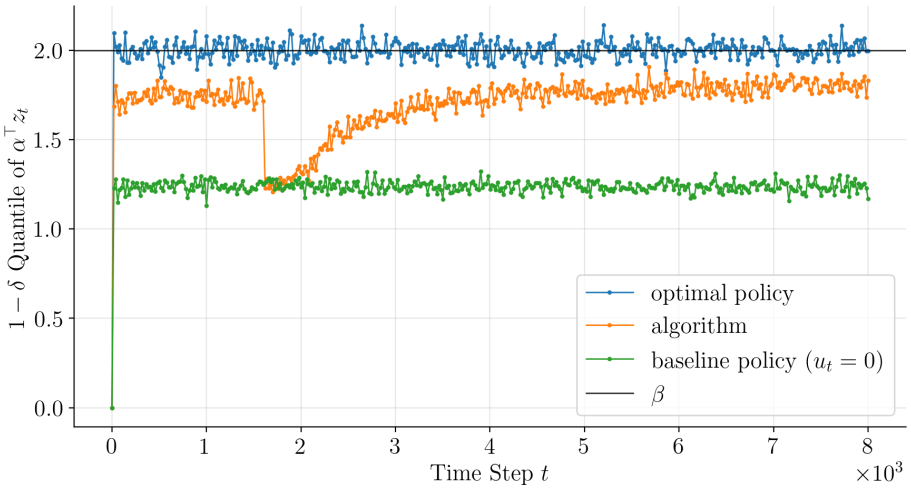
Numerical example

Two states, one input: require $\mathbb{P}(x_{t,2} \leq 2) \geq 0.9$.

Learning improves performance



Constraint quantile stays below limit

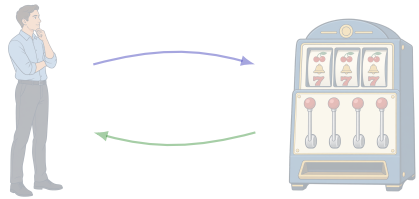


- The baseline is safe but conservative.
- The algorithm reduces regret relative to the baseline.
- A 0.9-quantile below 2 corresponds to the chance constraint.

Problem Settings for Safe Online Learning

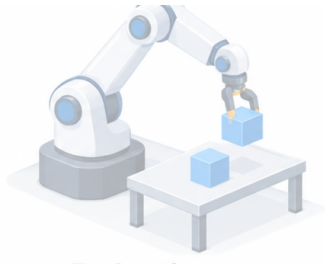
1. **Safe Stochastic Bandits**

   Learn unknown rewards and constraints from noisy feedback.



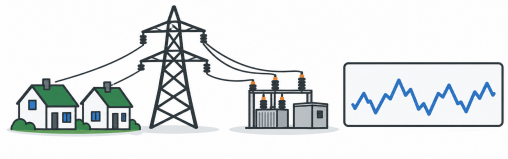
2. **Safe Online Control**

   Learn unknown dynamics while controlling the system.



3. **Safe Online Convex Optimization**

   Respond to changing costs while learning convex constraints.



Decisions in changing environments

How to make safe decisions when costs change *unpredictably* over time?







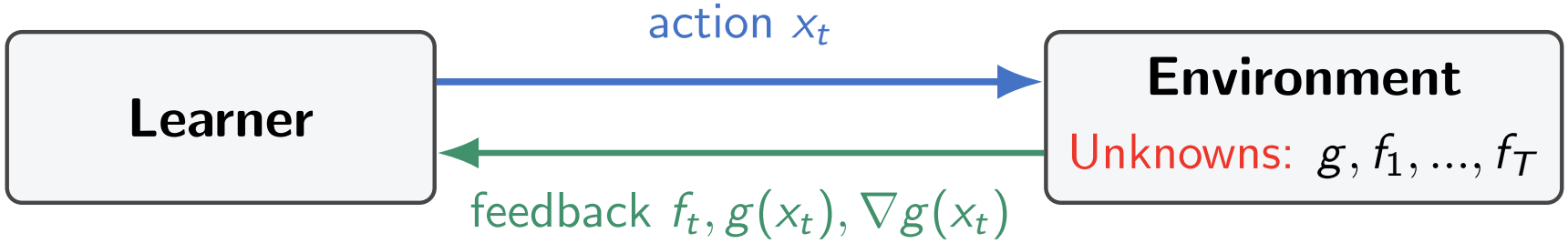
Costs may change **arbitrarily**.

Constraints must hold **at every step**.

Constrained online convex optimization with local feedback

Interaction at each round t:

1. Choose an action $x_t \in \mathbb{R}^d$.
2. Observe convex cost function f_t.
3. Observe local constraint feedback $g(x_t)$, $\nabla g(x_t)$.



Regret: realized cost compared to best choice in hindsight.

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \min_{x:g(x)\leq 0} \sum_{t=1}^{T} f_t(x)$$

Safety: $g(x_t) \leq 0$ for every t.

Gradient descent under constraints

Cost gradient step

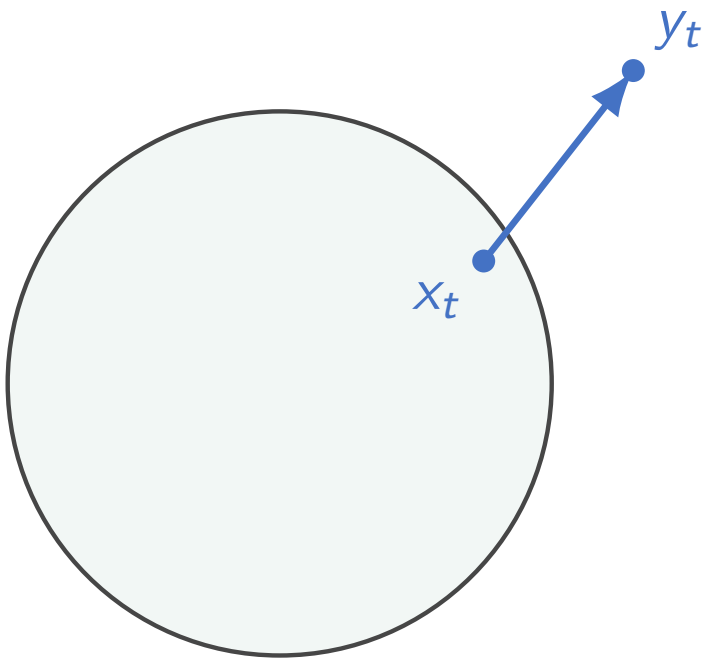
$$y_t = x_t - \eta \nabla f_t(x_t).$$

The step may leave the feasible set.

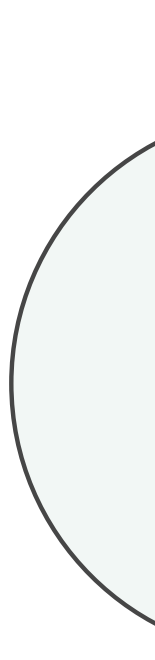
Traditional approach
Project back on to feasible set:

$$\Pi_{\mathcal{X}}(y_t) = \arg\min_{x \in \mathcal{X}} \|x - y_t\|$$

Requires full knowledge of constraint function!



$\mathcal{X} = \{x : g(x) \leq 0\}$



$\mathcal{X} = \{$

Our approach: apply a safety correction with *local constraint information*.

Local feedback defines a containing halfspace

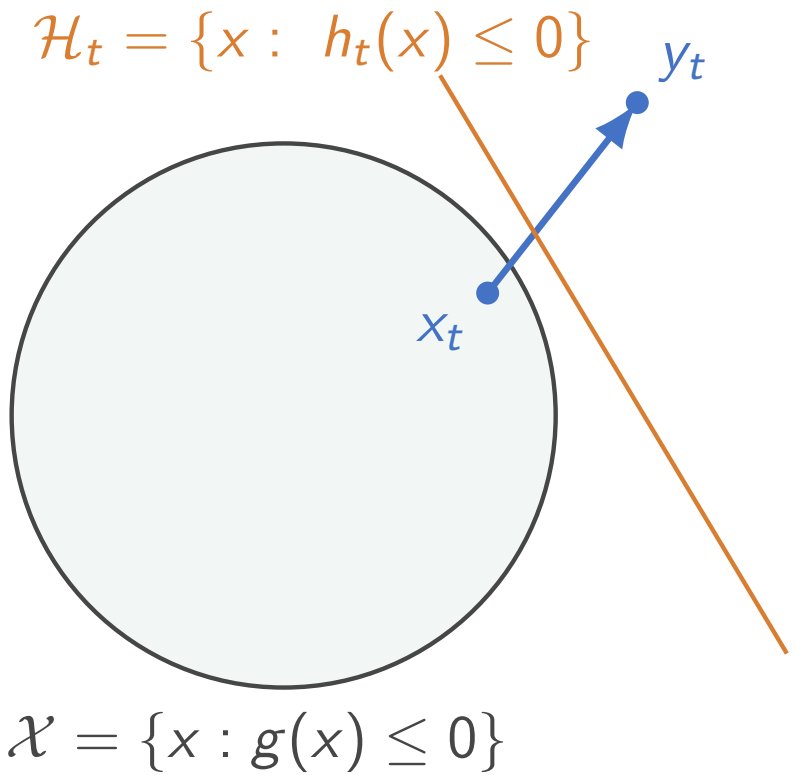
Convexity gives a global lower bound from local information:

$$g(x) \geq \underbrace{g(x_t) + \nabla g(x_t)^\top (x - x_t)}_{h_t(x)}$$

Halfspace formed by $h_t(x)$ contains feasible set:

$$\boxed{\mathcal{X} \subseteq \mathcal{H}_t},$$

$$\mathcal{H}_t = \{x : h_t(x) \leq 0\}.$$



A halfspace projection preserves progress

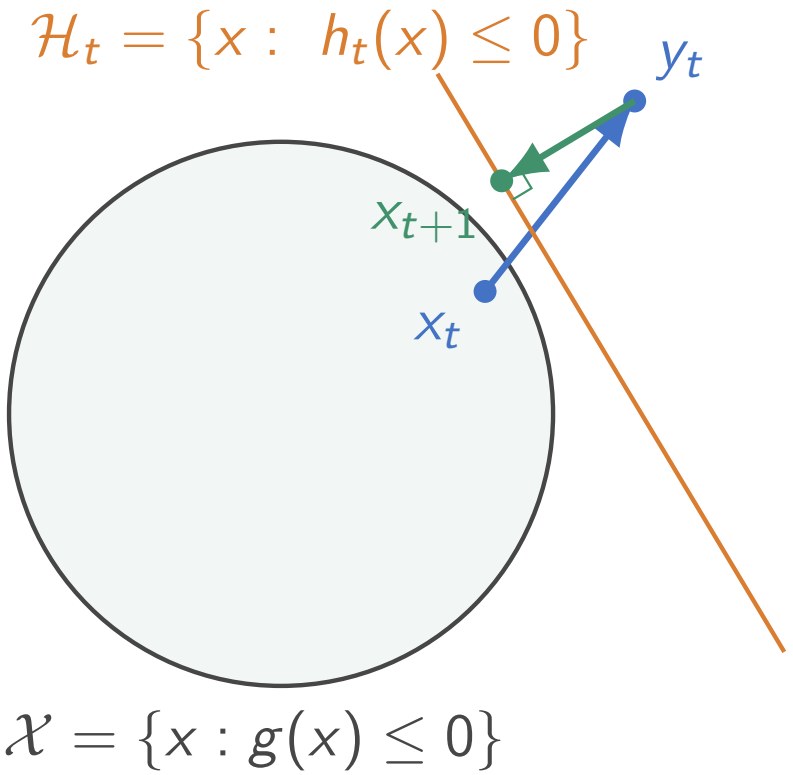
Halfspace projection:

$$x_{t+1} = \Pi_{\mathcal{H}_t}(y_t).$$

Does not increase distance to any feasible point:

$$\|x_{t+1} - x\| \leq \|y_t - x\|, \forall x \in \mathcal{X}$$

Implies low regret.



The halfspace contains infeasible points.

How can this correction ensure safety?

Halfspace projection maintains distance to feasible set

Lemma. Under constraint regularity, distance to feasible set satisfies:

$$\operatorname{dist}(x_{t+1}, \mathcal{X}) \leq \underbrace{\alpha \operatorname{dist}(x_t, \mathcal{X})}_{\text{contraction}} + \underbrace{\eta \|\nabla f_t(x_t)\|}_{\text{gradient step}}, \quad 0 \leq \alpha < 1.$$

$$\operatorname{dist}(x_t, \mathcal{X}) \leq \alpha^{t-1} \operatorname{dist}(x_1, \mathcal{X}) + \frac{\eta G_f}{1-\alpha} \qquad (\|\nabla f_t\| \leq G_f).$$

$$g(x_t) \leq G_g \alpha^{t-1} \operatorname{dist}(x_1, \mathcal{X}) + \frac{\eta G_f G_g}{1-\alpha} \qquad (\|\nabla g\| \leq G_g).$$

Tighten the constraint

Feed the *tightened* constraint to the algorithm: $g_\rho(x) = g(x) + \rho$.

$$g(x_t) \leq \underbrace{G_g \alpha^{t-1} \operatorname{dist}(x_1, \mathcal{X}_\rho)}_{\text{initial distance}} + \underbrace{\frac{\eta G_f G_g}{1-\alpha}}_{\text{gradient steps}} - \underbrace{\rho}_{\text{safety margin}}.$$

Choose $g(x_1) \leq -\rho$ and $\rho = \dfrac{\eta G_f G_g}{1-\alpha}$: $\boxed{g(x_t) \leq 0, \text{ for all } t.}$

Polyak feasibility steps: algorithm and guarantees

Algorithm:

1. Cost gradient step: $y_t = x_t - \eta \nabla f_t(x_t)$
2. Tightened halfspace projection: $x_{t+1} = \Pi_{\{x:\, h_t(x) \leq -\rho\}}(y_t)$

Theorem[6] With boundedness and constraint regularity, a strictly feasible start, and suitable $\eta, \rho \asymp T^{-1/2}$:

$$\boxed{R_T = O(\sqrt{T})} \quad \text{and} \quad \boxed{g(x_t) \leq 0 \quad \forall t}.$$

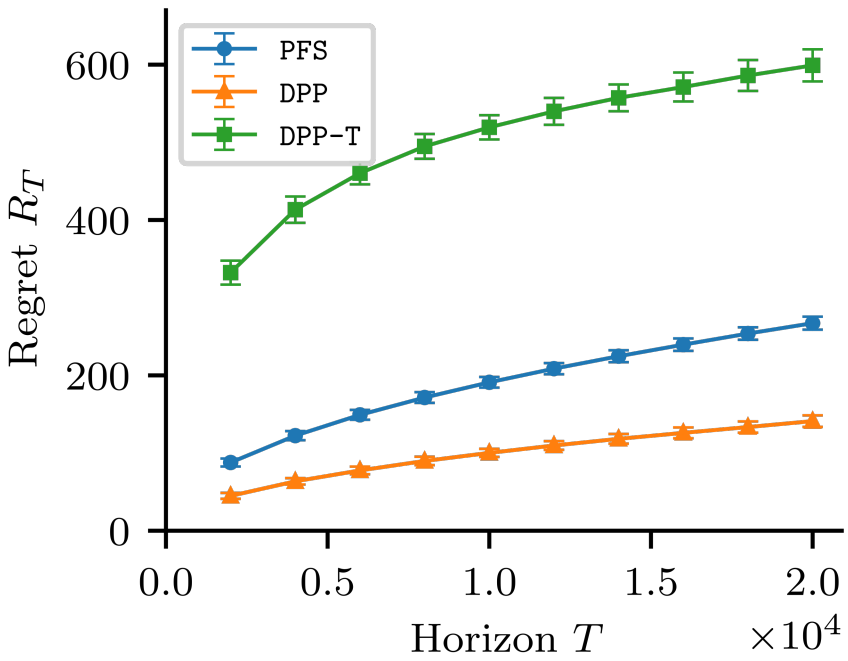
- $O(\sqrt{T})$ regret is **optimal** in online convex optimization.
- **First** algorithm for online convex optimization that **maintains constraint satisfaction** with only first-order feedback. Prior such works guarantee only $\sum_{t=1}^{T} g(x_t) = O(\sqrt{T})$.

[6] **S. Hutchinson** and M. Alizadeh, "Constrained Online Convex Optimization with Polyak Feasibility Steps," ICML 2025.

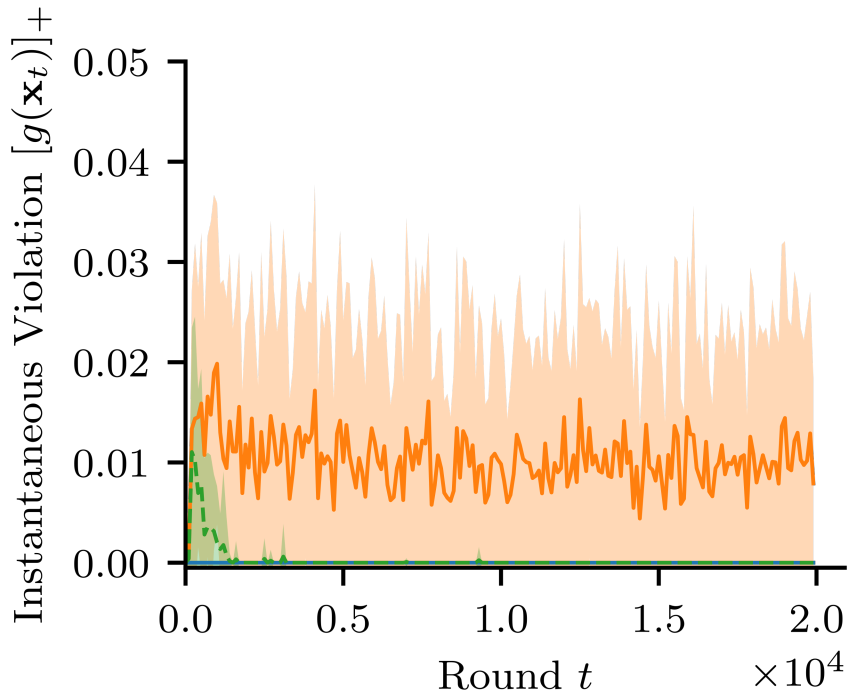
Numerical example

Changing quadratic costs over a fixed feasible region.

Regret



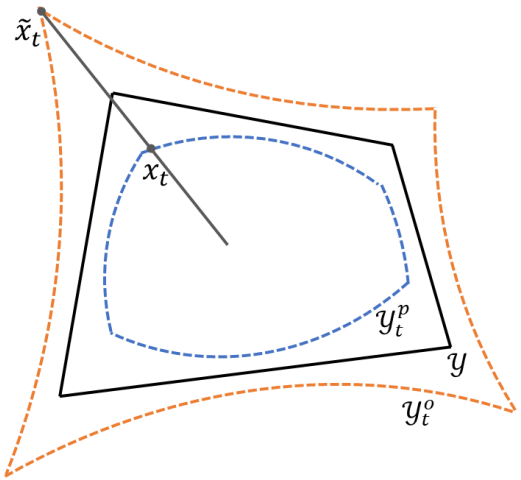
Violation at each round

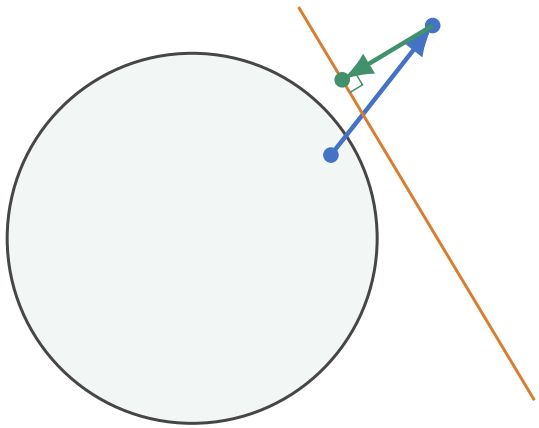


- Proposed algorithm (PFS) maintains feasibility while adapting to changing costs.
- The comparison methods illustrate the cost–feasibility tradeoff.

Conclusion

- **Safe Stochastic Bandits**
  - ROFUL algorithm: optimism + scaling
  - Matches state-of-the-art regret guarantees.
- **Safe Online Control**
  - Combined ROFUL with covariance-based policy synthesis.
  - First rate-optimal regret bounds for constrained LQR.
- **Safe Online Convex Optimization**
  - Gradient step with projection on to linearized constraint.
  - First algorithm with constraint satisfaction under first-order feedback.





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Thank You!